

Extension 1: (IDs ending with 3, 6, 9)

In this extension, you'll use your application to help a spaceship captain find the most efficient travel path between star systems. The star systems are represented as nodes in your app. In addition, certain star systems (nodes) contain special teleportation gates that allow instant travel between them at no cost. Add a new node type to your program to represent these teleportation hubs, and modify your shortest-distance path algorithm to incorporate this behavior.

Extension 2: (IDs ending with 1, 4, 7)

In this extension, you will use your application to analyze network connections. Each node in your application represents a server, and the paths between them represent network links. Add a new button to your application that assigns random colors to each connected network differently.

Extension 3: (IDs ending with 0, 2, 5, 8)

In this extension, you'll use your application to help a bus navigate its route. Add a new node type that carries a constant cost of 42. When computing the shortest path, if the bus visits this node, add the fixed cost to the total path length. Modify your shortest-distance path algorithm to incorporate this behavior.